Linear Algebra – MAT 2610

Section 1.4 (The Matrix Equation $Ax = b$)

Dr. Jay Adamsson

jay@aorweb.ca

jadamsson@upei.ca

Matrix Multiplication

Let A be an $m \times n$ matrix and v be a $n \times 1$ vector. The **product** Av is the linear combination of the columns of A using the corresponding entries in v as weights, and is given by :

$$Av = v_1 a_1 + \cdots + v_n a_n$$

Examples:

$$\underbrace{\begin{bmatrix} 1 & 4 \\ -4 & 0 \\ 3 & 2 \end{bmatrix}}_{3\times 2}\underbrace{\begin{bmatrix} 2 \\ -1 \end{bmatrix}}_{2\times 1} = (2)\begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} + (-1)\begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ -8 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 0 \\ 4 & -1 & 1 \\ -1 & 2 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x\begin{bmatrix} 2 \\ 4 \\ -1 \end{bmatrix} + y\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} + z\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2x + 4y \\ 4x - y + z \\ -x + 2y - z \end{bmatrix}$$

Theorem 3

If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$ and if b is in R^m, the matrix equation

$$Ax = b$$

Has the same solution set as the vector equation

$$x_1 a_1 + \cdots + x_n a_n = b$$

Which, in turn, as the same solution set as the system of linear equations whose augmented matrix is

$$\begin{bmatrix} a_1 & \dots & a_n & b \end{bmatrix}$$

Existence of Solutions

The equation $Ax = b$ has a solution if and only if b is a linear combination of the columns of A.

Example: is the following consistent for all values of b_1, b_2, b_3?

$$\begin{aligned} x + y + z &= b_1 \\ 2x + 3y &= b_2 \\ 2x + 2z &= b_3 \end{aligned}$$

$$x + y + z = b_1$$
$$2x + 3y = b_2$$
$$2x + 2z = b_3$$

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & b_1 \\ 2 & 3 & 0 & b_2 \\ 2 & 0 & 2 & b_3 \end{array}\right] \xrightarrow{2r_1 - r_2 \to r_2} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & b_1 \\ 0 & -1 & 2 & 2b_1 - b_2 \\ 2 & 0 & 2 & b_3 \end{array}\right]$$

$$\xrightarrow{2r_1 - r_3 \to r_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & b_1 \\ 0 & \boxed{-1} & 2 & 2b_1 - b_2 \\ 0 & 2 & 0 & 2b_1 - b_3 \end{array}\right] \xrightarrow{2r_2 + r_3 \to r_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & b_1 \\ 0 & \boxed{-1} & 2 & 2b_1 - b_2 \\ 0 & 0 & \boxed{4} & 6b_1 - 2b_2 + b_3 \end{array}\right]$$

$$\xrightarrow[\frac{1}{4}r_3 \to r_3]{-r_2 \to r_2} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & b_1 \\ 0 & \boxed{1} & -2 & -2b_1 + b_2 \\ 0 & 0 & 1 & \frac{2}{3}b_1 - \frac{1}{2}b_2 + \frac{1}{4}b_3 \end{array}\right] \xrightarrow{r_1 - r_2 \to r_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 3 & 3b_1 - b_2 \\ 0 & \boxed{1} & -2 & -2b_1 + b_2 \\ 0 & 0 & \boxed{1} & \frac{2}{3}b_1 - \frac{1}{2}b_2 + \frac{1}{4}b_3 \end{array}\right]$$

$$\xrightarrow{r_1 - 3r_3 \to r_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & b_1 - \frac{3}{2}b_2 - \frac{3}{4}b_3 \\ 0 & \boxed{1} & -2 & -2b_1 + b_2 \\ 0 & 0 & \boxed{1} & \frac{2}{3}b_1 - \frac{1}{2}b_2 + \frac{1}{4}b_3 \end{array}\right] \xrightarrow{r_2 - 2r_3 \to r_2} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & b_1 - \frac{3}{2}b_2 - \frac{3}{4}b_3 \\ 0 & \boxed{1} & 0 & -\frac{10}{3}b_1 + 2b_2 - \frac{1}{2}b_3 \\ 0 & 0 & \boxed{1} & \frac{2}{3}b_1 - \frac{1}{2}b_2 + \frac{1}{4}b_3 \end{array}\right]$$

There is always a solution for $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Theorem 4

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A, either they are all true statements or they are all false.

1. For each b in R^m, the equation $Ax = b$ has a solution.
2. Each b in R^m is a linear combination of the columns of A.
3. The columns of A span R^m.
4. There is a pivot position in every row of A.

Note: This theorem uses the coefficient matrix A, not the augmented matrix $[A \quad b]$

Computation of Ax

Row-Vector Rule for Computing Ax: If the product Ax is defined, then the ith entry in Ax is the sum of the products of corresponding entries from row i of A and from the vector x.

Example:

$$\underbrace{\begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & -1 \end{bmatrix}}_{2 \times 3} \underbrace{\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}}_{3 \times 1} = \begin{bmatrix} \square \\ \square \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1 \\ & & \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} (1)(1) + (1)(2) + (-1)(-1) \\ \_\_\_ \end{bmatrix} = \begin{bmatrix} 1 + 2 + 1 \\ \_\_\_ \end{bmatrix} = \begin{bmatrix} 4 \\ \_\_\_ \end{bmatrix}$$

$$\begin{bmatrix} - & - & - \\ 2 & 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} \_\_\_ \\ (2)(1) + (3)(2) + (-1)(1) \end{bmatrix} = \begin{bmatrix} \_\_\_ \\ 2 + 6 - 1 \end{bmatrix} = \begin{bmatrix} \_\_\_ \\ 7 \end{bmatrix}$$

Theorem 5

If A is an $m \times n$ matrix, u and v are vectors in R^n, and c is a scalar, then:

1. $A(u + v) = Au + Av$
2. $A(cu) = c(Au)$

Proof of Theorems 4 and 5 are in the textbook.

(1) $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \right) = \left(\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) + \left(\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \right)$

(2) $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \left(3 \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \right) = 3 \left(\begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} \right)$

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